



# Rising incidence of invasive bacterial infections in Belgium following the COVID-19 pandemic

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## Abstract

**Purpose:** To describe the evolution of invasive bacterial infections (IBIs) *Haemophilus influenzae* (Hi), *Streptococcus pneumoniae* (Spn), *Streptococcus pyogenes* (Spy) and *Neisseria meningitidis* (Nm), in Belgium over a 7-year time period (2017–2024). ; **Methods:** Case numbers and incidence rates were evaluated: pre-COVID (2017–2019), COVID (2020–2021) and post-COVID (2022–2024). ; **Results:** Incidence of IBIs was severely impacted during COVID in Belgium, respectively reductions of 45.1%, 59.1%, 69.8% and 80.5% for Hi, Spn, Spy and Nm, in 2020–2021 versus 2017–2019. Following lifting non-pharmaceutical interventions, IBIs increased towards end 2022. Spy and Hi infections decreased again in 2023–2024, however exceeding pre-COVID levels. Spn further increased in 2023–2024, while Nm is the only pathogen with lower cases in 2023–2024 than pre-COVID. ; **Conclusion:** While similarity for all IBIs was expected based on post-COVID immunity decrease, other factors like emerging bacterial clones and viral-bacterial interactions may have contributed.

**Keywords** Invasive bacterial disease · *S. pneumoniae* · *H. influenzae* · *S. pyogenes* · *N. meningitidis* · Incidence

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## Introduction

Invasive bacterial diseases (IBDs) are leading causes of global mortality and morbidity among all ages, however affecting mostly young children (<5 years old), adolescents and older adults (>65 years old). The most common causative agents of these life-threatening infections are *Streptococcus pneumoniae* (Spn), *Streptococcus pyogenes* (Spy), *Haemophilus influenzae* (Hi), and *Neisseria meningitidis* (Nm). All four bacterial pathogens can colonize the nasopharynx or throat of healthy individuals and are easily transmitted via respiratory droplets. While all four pathogens frequently cause non-invasive diseases, more severe manifestations (i.e. bacteraemia, sepsis, empyema and meningitis) are described when the pathogen is detected in normal sterile body compartments, indicating invasive disease. Mandatory notification has been installed in Belgium for three of the four pathogens, namely invasive group A streptococcal infections (iGAS), Hi type b (Hib) and Nm [1]. While all four IBDs can be treated in first-line with beta-lactam antibiotics, three pathogens are ranked on the WHO bacterial priority pathogens list of 2024 [2] due to the occurrence and emergence of antimicrobial resistance.

Except for *Spy*, prevention of invasive bacterial infections (IBIs) has been implemented since more than 15 years within the national childhood immunisation program in Belgium – free of charge for *Hib* since 1993, *Nm* (currently serogroups ACWY) since 2003 and *Spn* (currently 13 serotypes) since 2007 – characterized by a high and stable vaccine uptake in young children [3]. Epidemiological changes have been evaluated regularly throughout time, in the context of implementation of vaccination whenever applicable, however parallel trends for all four IBIs have not yet been reported for Belgium. Here, we aim to describe in detail the evolution of all four IBIs in Belgium over a 7-year time period (2017–2024).

## Materials and methods

Epidemiological surveillance of all four bacteria is performed by four national reference centres (NRCs) in Belgium [4], hosted at four different institutes (*Spy* at University Hospital Antwerp, *Spn* at University Hospitals Leuven, *Nm* at Sciensano and *Hi* at University Laboratory Brussels), each focusing on the national epidemiology of the respective IBIs since at least 2011. Coverage of these voluntary laboratory-based surveillance programs remained largely stable over the last years (calculated based on blood culture practices of the participating laboratories and/or mandatory notification) and ranges depending on the NRC from 70% to more than 90% [5–8], providing accurate insights in the evolution of IBIs over a long time period in Belgium. Bacterial isolates from a stable number of clinical laboratories located across the country are sent to the respective

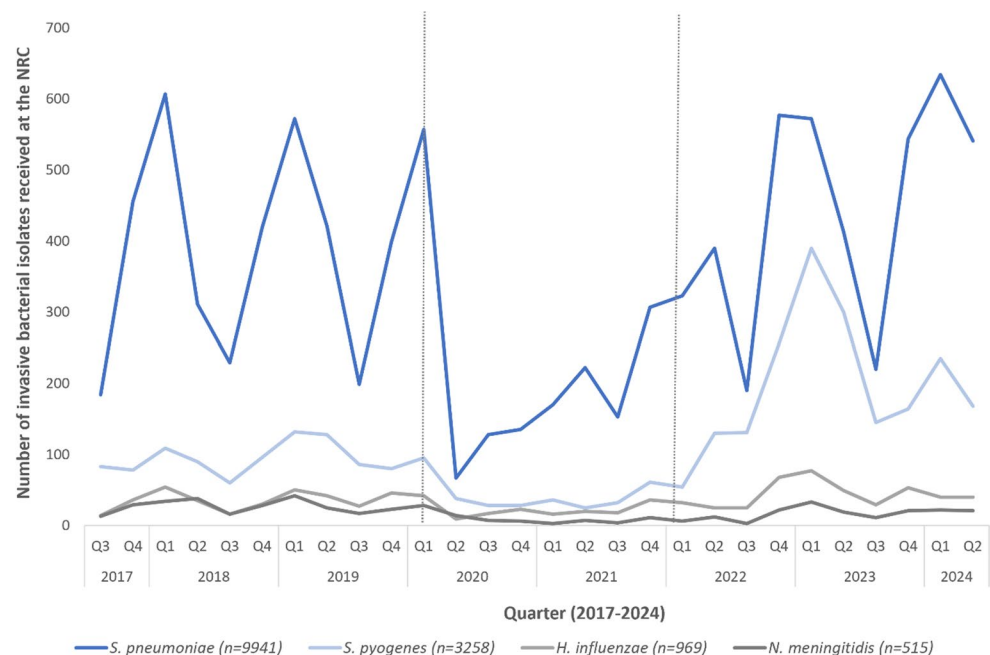
NRCs for identification, typing, antimicrobial susceptibility testing and/or in-depth characterisation. Case numbers of invasive infections were shared among the four respective NRCs for epidemiological years (July to June) 2017–2018 to 2023–2024, distinguishing between children (<18 years) and adults (≥18 years). The evolution of case numbers and incidence rates (corrected based upon uniform coverage estimates for each of the respective surveillance programs – per 100,000 individuals; using population denominators as available on Statbel [9]) of IBIs was evaluated over time, identifying three periods: pre-COVID (two epidemiological seasons: 2017–2019), COVID (epidemiological season 2020–2021) and post-COVID (two epidemiological seasons: 2022–2024). Cases with unknown age were not considered for the age-specific analyses. Despite uniform and high coverage of all four IBI surveillance programs, potential residual bias from non-participating laboratories cannot be excluded.

## Results and discussion

### Drastic reduction of IBIs during the COVID-19 pandemic followed by post-COVID rise

Measures taken to prevent transmission of SARS-CoV-2 (such as hand washing, face masking, social distancing) also heavily disturbed transmission of other respiratory pathogens and prevalence of bacterial superinfections, accounting for an overall reduction of 45.1%, 59.1%, 69.8% and 80.5% in the yearly number of invasive infections with *Hi*, *Spn*, *Spy* and *Nm* respectively (Fig. 1), from pre-COVID

**Fig. 1** Number of invasive isolates received at the national reference centres (NRCs) per quarter for the epidemiological years 2017–2018 to 2023–2024, respectively for *S. pneumoniae*, *S. pyogenes*, *H. influenzae* and *N. meningitidis*. Three time periods were considered: pre-COVID (2017–2019), COVID (2020–2021) and post-COVID (2022–2024). Vertical lines mark the estimated start and end date of the pandemic period of restrictions as calculated for Belgium in [12]



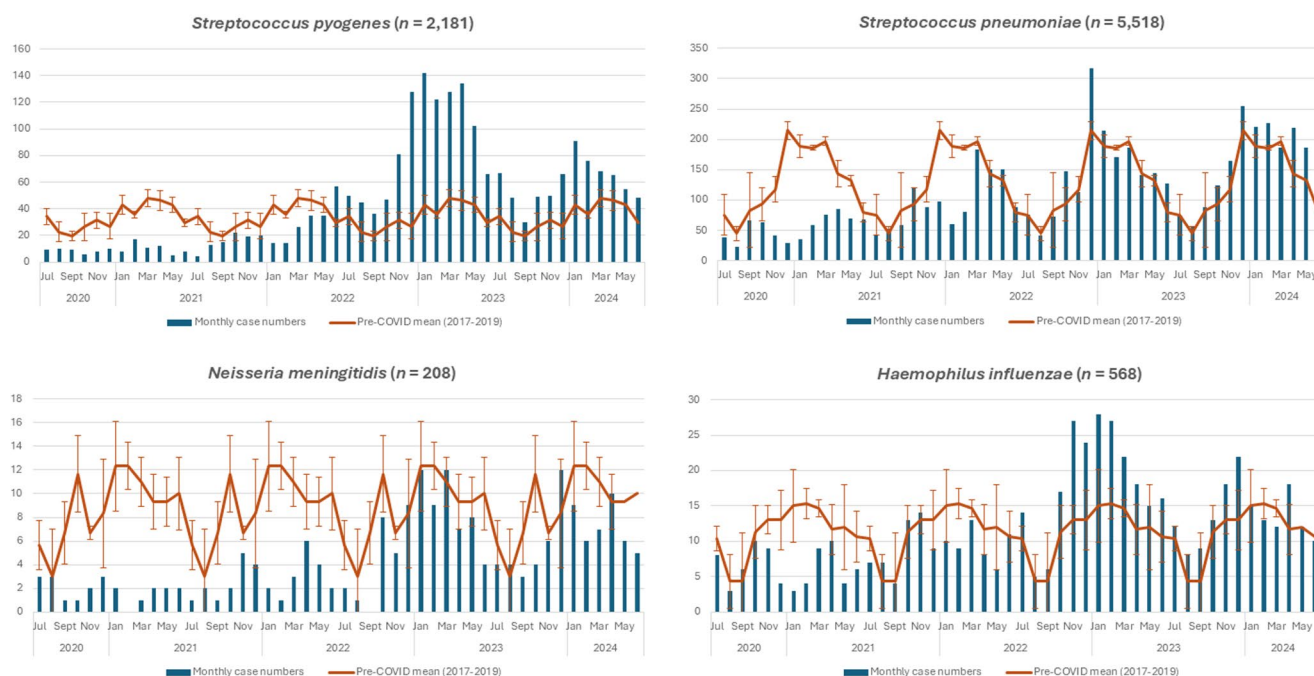
(2017–2019) to COVID (2020–2021), in agreement to other countries across the world [10–11]. In comparison to neighbouring countries, primary schools were relatively soon reopened in Belgium (partial reopening in May–June 2020), while restrictions were prolonged in nursing homes.

Following lifting of non-pharmaceutical interventions post-COVID in January 2022, an increase in all IBIs was noted towards the end of 2022 in concomitance with a rise in viral respiratory infections [13]. In agreement with other countries across Europe, exceptional high numbers of group A streptococcal infections were noted for epidemiological year 2022–2023 in Belgium, resulting in an increase of cases of 64% compared to pre-COVID years. The upsurge in iGAS infections in Belgium coincided with the predominance of *emm1 Spy*, more specifically the toxigenic M1<sub>UK</sub> lineage [14–15]. Compared to pre-COVID years 2017–2019, 36.8% and 8.7% more infections with *Hi* and *Spn* respectively were observed in 2022–2023, while a decrease of 31.9% was seen for *Nm* between 2017–2019 and 2022–2023.

Consistently for all four IBIs, infections drastically decreased in 2020–2021, followed by a parallel rise in case numbers in 2022–2023. In the second post-COVID epidemiological year 2023–2024 however, the four IBIs were impacted differently, expressed in three distinct scenarios.

In the first scenario, corresponding to both *Spy* and *Hi*, infections decreased again (−33.9% and −26.0% compared to post-COVID year 2022–2023). However, despite reduction in cases for both bacterial pathogens in 2023–2024 compared to 2022–2023, counts still largely exceeded those of pre-COVID years 2017–2019, respectively +45.5% and +14.5% (Fig. 2). The second scenario concerns a stabilization for both post-COVID years 2022–2024, as for *Nm* no large differences were observed (−2.6%), making it the only pathogen post-COVID to still have lower case numbers compared to pre-COVID years (−33.6%). *Spn* makes up the third scenario, with continued increased case numbers observed post-COVID, respectively +10.7% from 2022–2023 to 2023–2024, reflecting an increase of nearly 18% compared to pre-COVID levels (Table S1).

Each of the four bacterial pathogens is characterized by a seasonal pattern as depicted by the pre-COVID (2017–2019) case numbers in Fig. 2, however clearly disturbed during the COVID-19 period for which non-pharmaceutical interventions in the context of the COVID-19 pandemic were in place (March 2020 – January 2022 [12]). Along the recovery in the monthly number of IBIs post-COVID, seasonality is restored since epidemiological year 2023–2024, with the exception of *Nm* for which deviations are still seen.



**Fig. 2** Monthly case numbers for *S. pyogenes*, *S. pneumoniae*, *N. meningitidis*, and *H. influenzae* for epidemiological years 2020–2021 to 2023–2024, plotted in blue bars. Pre-COVID (2017–2019) mean case

numbers are visualised in orange lines, together with their respective standard deviations ( $\pm 1$  SD) as calculated for the number of infections between 2017 and 2019 for each respective bacterial pathogen

## Difference in incidence of IBDs between children and adults

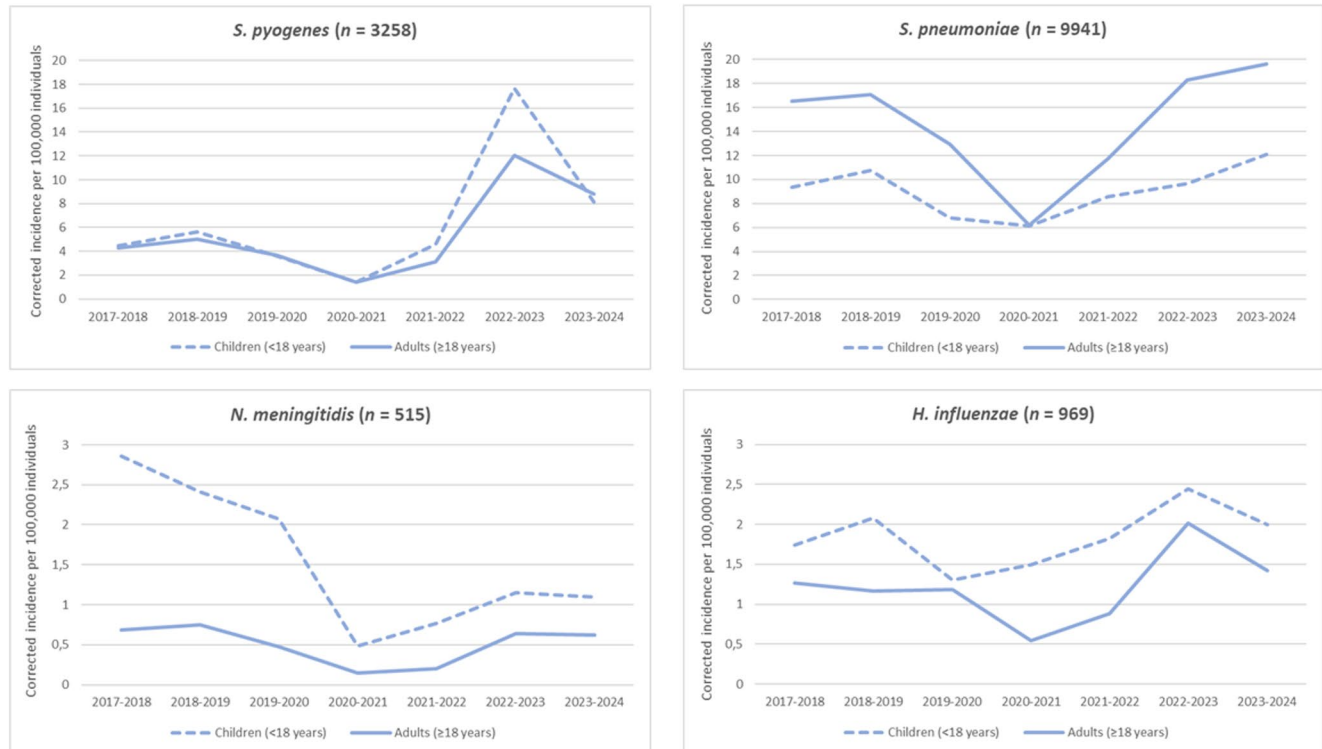
Installing and lifting of COVID-19 containment measures differently impacted incidence rates of IBDs for children (< 18 years) and adults (≥ 18 years). For the COVID year 2020–2021, incidence rates per 100,000 individuals decreased similarly for children and adults for *Spn*, while the large increase in infections that was observed post-COVID, was clearly driven by children (increase of incidence rates of 1.42 to 17.68 per 100,000 children; Fig. 3 and Table S1). Likewise, the decrease in incidence rates between 2022–2023 and 2023–2024 was more pronounced in children for *Spn* (17.68 to 8.13 per 100,000 children). For *Hi*, a different scenario was observed compared to *Spn*, with a more pronounced decrease in adult cases compared to children (−38.9% versus −22.0%) following implementation of non-pharmaceutical interventions. Similarly, both the re-increase between 2020–2021 and 2022–2023 and the decrease between 2022–2023 and 2023–2024 in *Hi* incidence rates was higher in adults compared to children.

Also for *Spn*, adults seemed to be more impacted throughout the studied period, with higher drops and overshoots for adults compared to children for both the COVID (16.80 to 6.20 per 100,000 adults) and post-COVID period (6.20 to 19.63 per 100,000 adults). Current continued increasing

*Spn* case numbers are still mainly due to the high number of invasive pneumococcal infections in (older) adults [16]. In contrast, *Nm* incidence rates in children were most impacted by the containment measures installed during the COVID-19 pandemic (Fig. 3). The re-increase in *Nm* case numbers post-COVID resulted to incidence rates comparable to pre-COVID in adults. In contrast, *Nm* incidence in children remained far below pre-COVID levels.

## Importance of long-term continued surveillance programs

Long-term surveillance programs of IBD clearly show the impact of preventive measures of the COVID-19 pandemic in Belgium, in line with earlier reports of reduced case numbers for all four bacterial pathogens, while followed by re-stabilization and overshooting of infections for *Spn*, *Spn* and *Hi* [10–13]. For all three pathogens, during the epidemiological year 2023–2024 higher case counts were observed in comparison to pre-COVID levels. While *Nm* was the most impacted bacterial pathogen with a reduction of over 80% in case numbers during the COVID-19 pandemic, post-COVID it is the only pathogen for which still a lower number of infections is recorded compared to pre-COVID years. While similar patterns were expected based on anticipated



**Fig. 3** Incidence rates per 100,000 individuals for *S. pyogenes*, *S. pneumoniae*, *N. meningitidis*, and *H. influenzae* for epidemiological years 2017–2018 to 2023–2024, corrected for the coverage of the respective

surveillance programs. Incidence rates are visualised for both children (< 18 years) and adults (≥ 18 years)



post-COVID immunity decrease, all four bacteria were impacted differently, as well as the affected age groups, showing the potential contribution of other factors such as impact of different viral-bacterial interactions [13], transmission modes, carriage rates [17], and waning immunity or limitations of vaccination coverage, warranting further investigation. The sustained decline in *Nm* is consistent with the continued impact of the installed ACWY vaccination programme, strengthened by build-up herd effects. The continued rise in *Spn* infections, most pronounced in (older) adults, likely reflects serotype dynamics and age-related susceptibility [16]. The more modest and partial rebound above pre-COVID levels for *Hi* and *Spy* may relate to pandemic-induced shifts in population immunity, while for *Spy* the emergence of (new) bacterial clones [14–15] has been reported to play a key role. Interestingly, seasonal patterns for all pathogens were almost immediately restored in 2022 after the end of containment measures. Continued surveillance of IBIs in later post-COVID epidemiological seasons will be key to evaluate whether pre-COVID levels will be restored or further exceeded.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s10096-025-05334-y>.

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**Author contributions** All authors contributed to the study conception and design. Data collection was performed by Lize Cuypers, Stefanie van Kleef – van Koeveringe, Veerle Matheussen, Wesley Mattheus and Delphine Martiny. Analysis was conducted by Lize Cuypers. The first draft of the manuscript was written by Lize Cuypers and Stefanie Desmet. All other authors commented on pre-final versions of the manuscript. All authors read and approved the final manuscript.

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**Data availability** The raw case numbers and incidence rates for all four bacterial pathogens are provided as total, as well as for the two studied age groups (children and adults) per epidemiological year studied, in Supplementary material (Table S1).

## Declarations

**Ethics approval** All four institutes are appointed as national reference centre for the respective invasive bacterial infections, for which the role to perform epidemiological surveillance is described in the Royal Decree of February 9, 2011.

**Competing interests** The authors declare no competing interests.

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